

CLONED_KM Kulim

PART A:

1.

a. 2

b. $H = \frac{3}{4}, K = \frac{17}{4}$

2.

a. $-\frac{3}{4x^{3/2}}$

b. shown

3. $\frac{5}{3} \text{cms}^{-1}$

PART B

1. $z + w = \frac{3}{2} - \frac{1}{2}i = \frac{\sqrt{10}}{2} [\cos(-0.322\text{rad}) + i \sin(-0.322\text{rad})]$

2.

a. $\left(2, \frac{7}{2}\right]$

b. $\left\{x : x < -8, x > -\frac{8}{3}\right\}$

3.

a. $p^{-1}(x) = \frac{-\ln x}{2}$

b. $1 + e^{-6x}$

c. -1.145

4.

a.

i. 1

ii. $f(x)$ is continuous at $x = 0$

b. 2

5.

a. $\frac{3}{(3+x)(3-x)}$

b. 1 or $1/4$

6.

a. $-t^2 - 2t$

b. 16

7.

a. (3, 6) relative minimum, (-3, -6) relative maximum

b.

i. Shown

ii. 64